

PINESPHERE STAY

Staff Management Module

Logic, Workflow, Implementation Plan & Database Blueprint

Source Document	PineStay — Business Requirement Document (BRD)
Module in Scope	Staff Management (V1)
Platform	Mobile-first, Offline-First Property Management System
Tech Stack	Flutter (Android first) · FastAPI · PostgreSQL (Cloud) · SQLite / ObjectBox (Local)
Document Type	Module Blueprint — Logic, DB Design, Workflow & Dashboards

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1. Module Overview & Objectives

The Staff Management module governs every employee working at a property on the Pinesphere Stay platform — from the property Owner down to Housekeeping and Security staff. It sits inside the **Property Owner App** (Flutter, offline-first) and is one of the modules that must continue to function fully without internet, because staff attendance, task assignment and role enforcement happen throughout the day on the ground, regardless of connectivity.

Per the BRD, this module is directly responsible for: **Attendance, Leave, Salary, Roles, Performance and Task Assignment** (Page 10 — Staff Management), and it feeds the Role-Based Permission engine referenced in the Security section and the User Hierarchy diagram (Owner → Manager / Reception / Accountant → Housekeeping / Maintenance / Restaurant → Guest).

The module has four responsibilities:

- **Identity & Role Management** — create staff profiles, assign one of the BRD-defined roles (Owner, Manager, Reception, Cashier, Housekeeping, Maintenance, Restaurant, Security, Auditor, Accountant) and bind granular module-level permissions to each role.
- **Day-to-day Operations** — attendance capture, leave application/approval, task assignment (e.g. a housekeeping task tied to a room, or a maintenance complaint), and performance review.
- **Payroll Support** — monthly salary computation, payment status tracking and salary history, feeding the Accountant role and the Expense/Accounting module.
- **Governance** — every create/update/delete/login action by staff is written to an immutable audit log, and every staff device is bound and controllable from the Device Management module and Super Admin's Remote Device Control.

Design Principle: Staff Management must obey the same offline-first contract as the rest of the app — every read and write happens against the local SQLite/ObjectBox store first, and is queued for the Synchronization Engine to push/pull once internet is detected.

2. Staff Roles Defined in the BRD

The BRD explicitly names the following operational roles under a property (Page 26 — Staff Management, Page 18 — User Hierarchy, Pages 27–29 — Owner/Staff Login permissions):

Role	Scope (as per BRD)
Owner	Single primary owner per property. Full control: create staff, change prices, manage subscription, manage devices, view all reports, approve refunds, manage settings.
Manager	Operational authority delegated by Owner; broad access across bookings, rooms, staff and reports (per Role Matrix, Page 32-33).
Reception	Create booking, check-in, check-out, collect payments, view rooms. Cannot delete data, change subscription, manage users or export data.
Cashier	Payment collection focused role (referenced alongside Reception/Accountant in staff role list).
Housekeeping	Cleaning, laundry, room status updates, lost & found.
Maintenance	Complaint tracking for furniture, electrical, plumbing, AC, TV, fan.
Restaurant	Order taking, kitchen status, billing, stock/inventory deduction (Addon module).
Security	Named as a staff role in the BRD; scoped for premise/guest safety oversight.
Auditor	Named as a staff role in the BRD; read-focused, aligned to Audit Log visibility.
Accountant	Payments, expenses, GST, reports, invoices.

Design decision: roles are modeled as data (a `roles` table), not hard-coded enums. The ten roles above are seeded as *system roles* at property creation time, but the Owner can still create custom roles later — this keeps the module flexible for the Enterprise/Custom subscription plans mentioned on Pages 15–16.

3. Core Logic Design

3.1 Role & Permission Enforcement Logic

Every module screen and API endpoint checks a permission tuple (role_id, module_name, action) before allowing an action. This permission set is downloaded and cached locally on first login so checks work fully offline. The Owner login is hard-restricted: only one staff row per property may have is_primary_owner = true.

3.2 Attendance Logic

Staff punch in/out using Biometric, PIN or Manager-marked entry (Manual). A staff member can have only one attendance record per calendar date (enforced by a unique constraint on staff_id + date). If no check-in exists by end of business day and the day is not on approved leave, a background job (running locally, then reconciled on sync) marks the day **Absent**. Half-day is derived when check-in and check-out span less than a configurable minimum working duration.

3.3 Leave Management Logic

A leave request moves through Pending → Approved/Rejected, approvable only by a role holding can_approve on the Staff module (typically Owner/Manager). Leave balance is computed on demand: allocated days (from leave_types.max_days_per_year) minus the sum of approved leave days for the current year. Approved leave dates automatically create corresponding attendance rows with status **On-Leave** so the Attendance Dashboard never shows a false Absent.

3.4 Salary Logic

Net salary is computed as: Basic + Allowances + Overtime – Deductions – Advance Adjusted. One salary record is allowed per staff, per month, per year (unique constraint). Salary generation is restricted to Owner/Accountant. Payment status transitions Pending → Partially Paid → Paid, and every status change is captured in the audit log because it is a financial action.

3.5 Task Assignment Logic

Tasks can originate from another module (e.g. a Housekeeping task auto-created at guest check-out, or a Maintenance complaint logged against a room) or be created manually by a Manager. Each task stores an optional related_module and related_record_id so it can deep-link back to the originating room/booking/complaint. Overdue tasks (due_date passed, status still Pending/In-Progress) are automatically flagged for the Owner/Manager dashboard.

3.6 Performance Logic

Performance is captured as periodic reviews (rating 0–5) tied to a review window, plus two derived, system-calculated signals that feed the AI daily summary mentioned in the BRD (Page 15): attendance consistency (% present) and task completion rate (% completed on time). These derived metrics are computed at read-time from staff_attendance and staff_tasks — they are not stored redundantly.

3.7 Staff–Device Binding Logic

A staff member logs in from a registered device. A session row is created on login and closed on logout; the currently active session determines whether the Super Admin's Remote Device Control (Page 25–26) can force-logout that staff member instantly on next connectivity. This directly reuses the Device Management module rather than duplicating device data.

3.8 Offline Sync & Conflict Resolution Logic

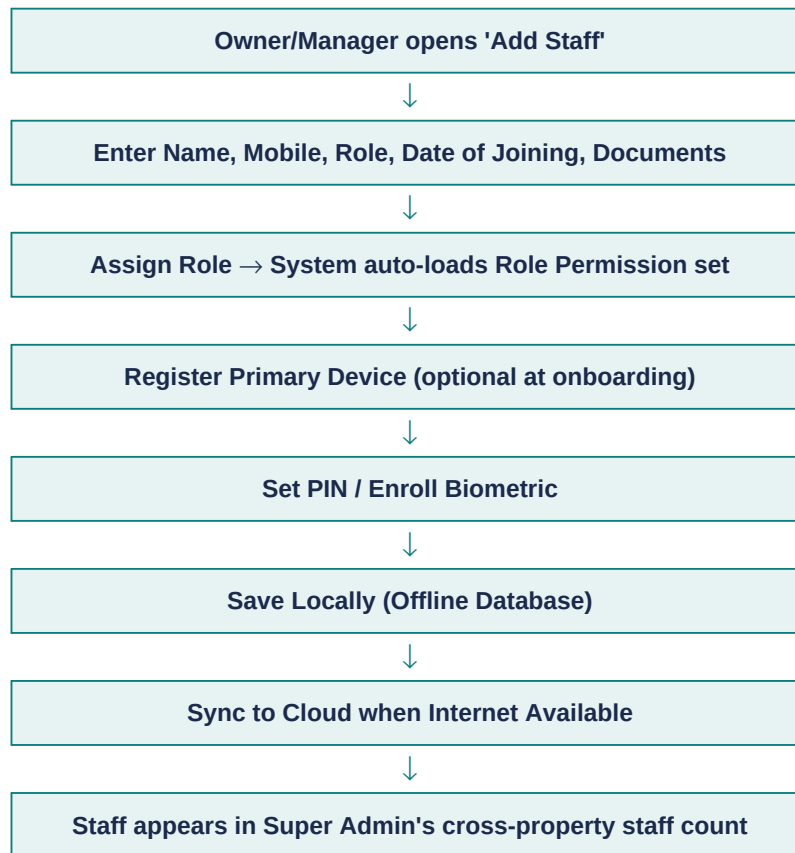
Every staff-related row carries `version`, `updated_at` and `sync_status`. On reconnect, the Sync Engine (Page 4 of the BRD) pushes local changes and pulls cloud changes. Non-critical fields (e.g. remarks, photo) use last-write-wins by `updated_at`. Financially or operationally sensitive fields — salary amounts, role changes, permission changes, leave approval status — are flagged **conflict** instead of auto-merged, and surfaced to the Owner for manual resolution, in line with the BRD's explicit 'Conflict Resolution' step in the sync flow (Page 3–4).

3.9 Audit Logging Logic

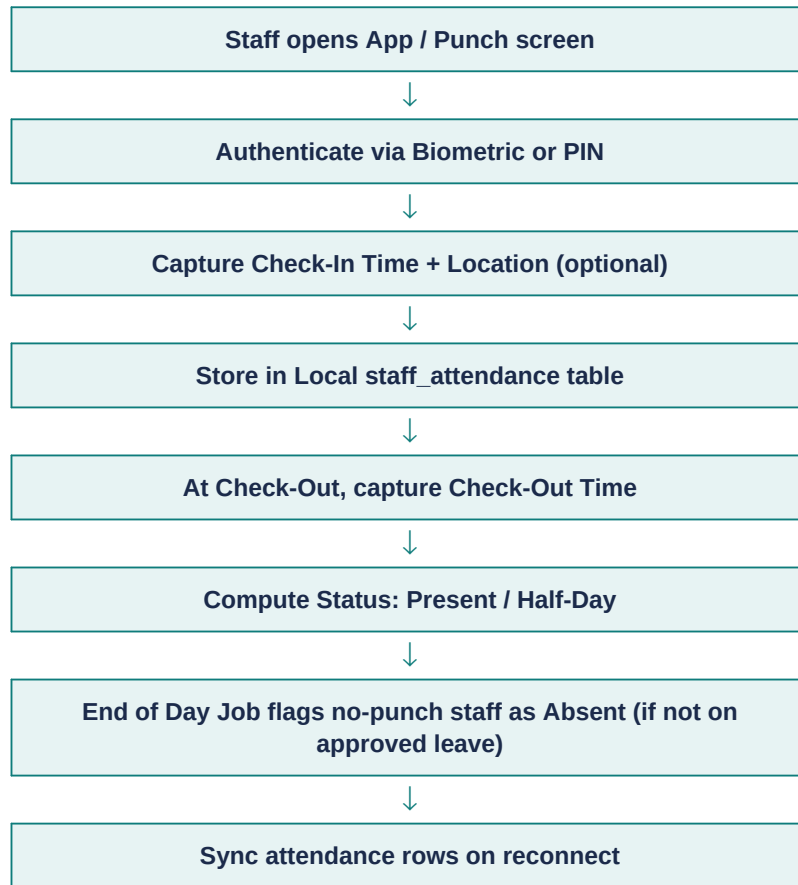
Every create, update, delete, login, approval and rejection on a staff-related entity writes one immutable row to `staff_audit_log`, storing `old_value/new_value` as JSON snapshots. The BRD is explicit: "Nothing can be deleted from the audit log" (Page 31) — so this table has no delete permission at the application layer, and destructive SQL operations are blocked at the database-role level too.

4. Workflows

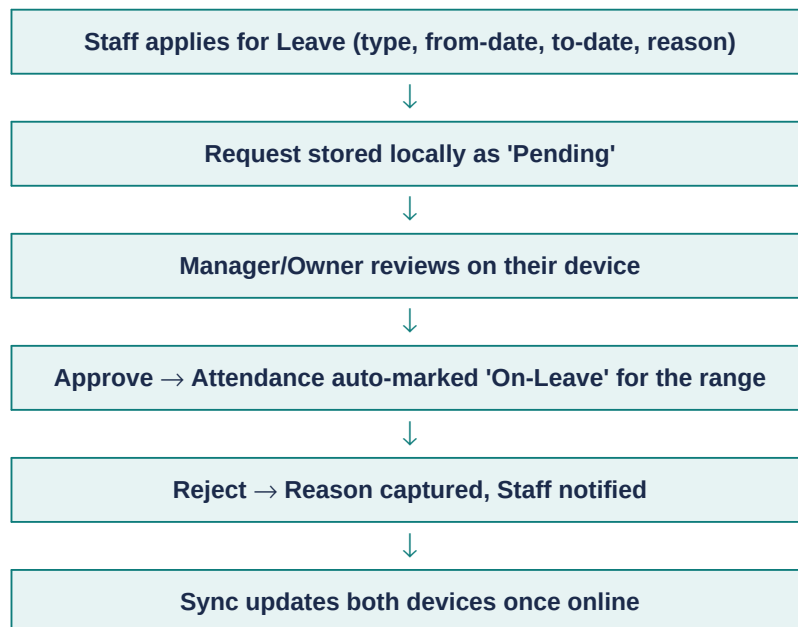
4.1 Staff Onboarding Workflow



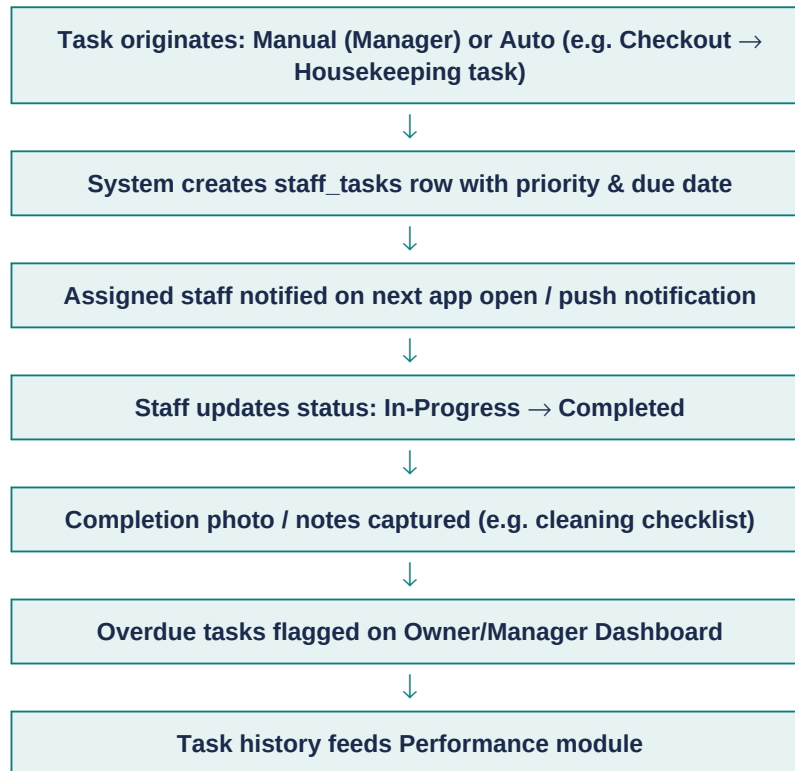
4.2 Daily Attendance Workflow



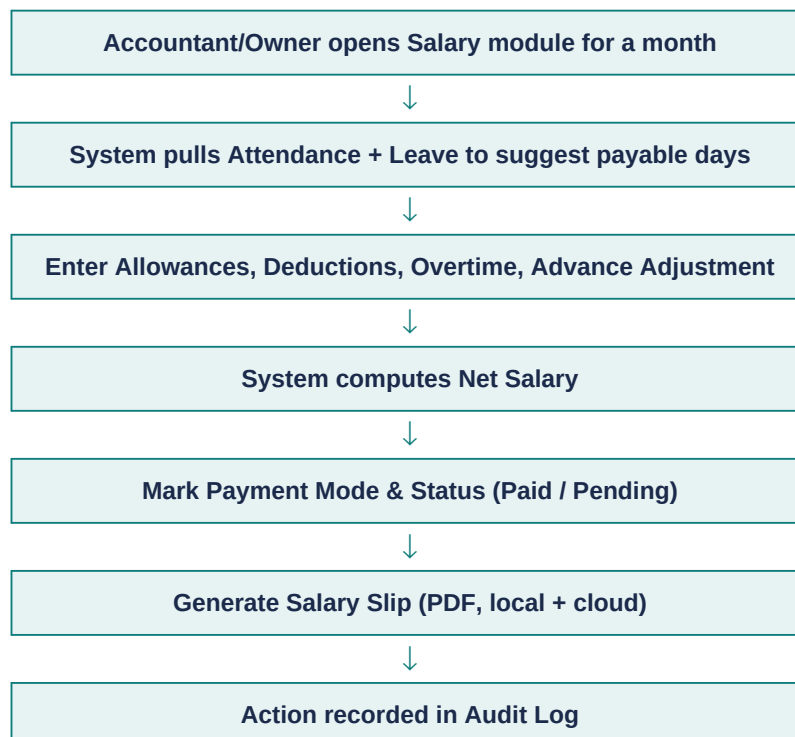
4.3 Leave Application & Approval Workflow



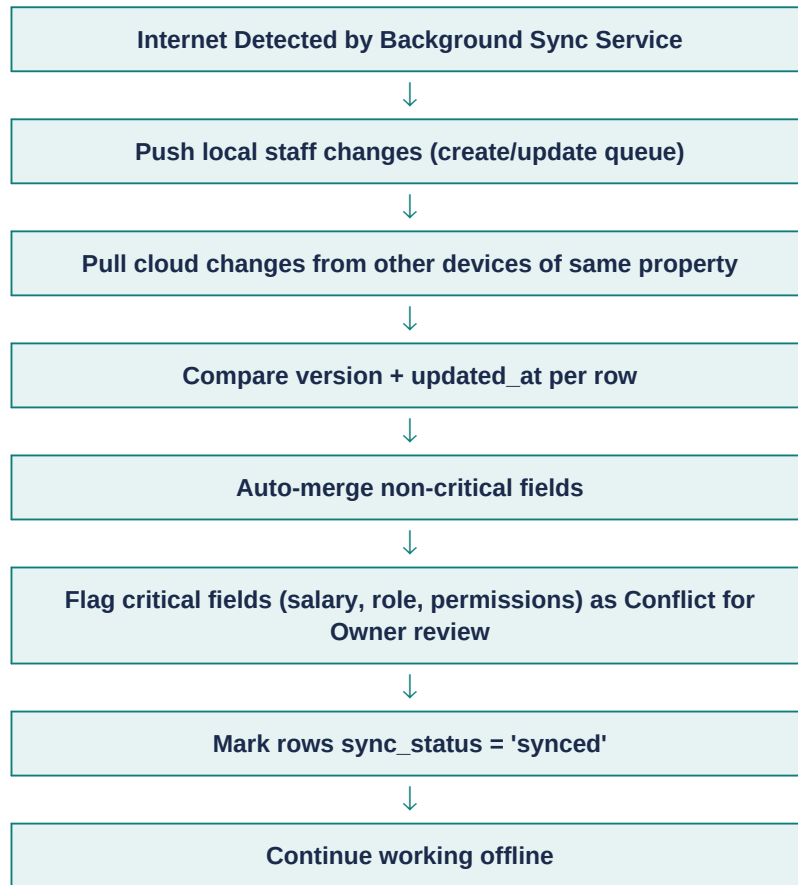
4.4 Task Assignment Workflow



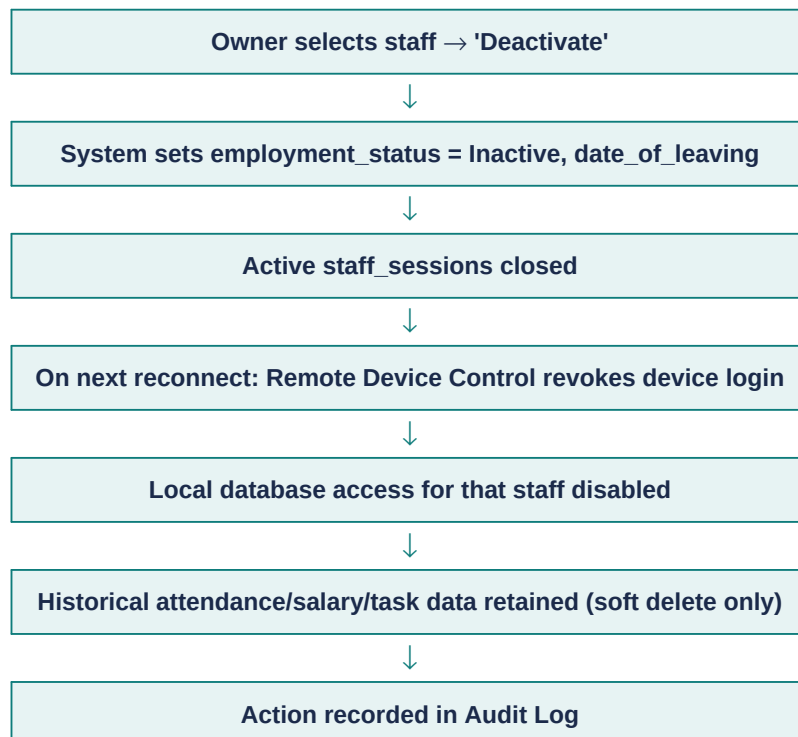
4.5 Monthly Salary Workflow



4.6 Offline → Online Sync Workflow (Staff Data)



4.7 Staff Deactivation / Offboarding Workflow



5. Implementation Plan (Phase-wise, no timelines)

Phases are ordered by dependency, not by date — build top to bottom.

Phase	Focus	Deliverables
Phase 1	Foundation	roles, role_permissions tables · seed the 10 BRD system roles · basic staff CRUD (offline-first) · Owner single-primary constraint.
Phase 2	Authentication & Device Binding	PIN/Biometric login for staff · staff_sessions table · integration hook with Device Management module.
Phase 3	Attendance	Punch in/out, manual marking by Manager, auto-absent job, attendance calendar view.
Phase 4	Leave Management	leave_types seed data, apply/approve/reject flow, leave balance calculation, auto-sync with attendance.
Phase 5	Task Assignment	Manual + auto-generated tasks, linkage to Housekeeping/Maintenance/Restaurant modules, overdue flagging.
Phase 6	Salary & Payroll	Salary computation engine, payment status tracking, salary slip generation.
Phase 7	Performance	Manual review entries + derived metrics (attendance %, task completion %).
Phase 8	Documents & Verification	Upload ID proofs, verification flag, secure object storage sync.
Phase 9	Audit Logging	Immutable logging hooked into every write operation across the above phases.
Phase 10	Dashboards & Reports	Owner/Manager dashboard, Staff Directory, Attendance dashboard, Self-service dashboard, Super Admin analytics.
Phase 11	Sync Hardening	Conflict resolution UI for critical fields, retry logic, data compression, encrypted sync (aligns with BRD Sync Engine, Page 4).

Each phase should be built local-first (SQLite/ObjectBox + Flutter UI) before wiring the corresponding FastAPI endpoint and sync payload — this keeps the module usable offline even mid-development.

6. Database Design — Entities & Relationships

The Staff Management module introduces 12 entities. All tables exist both in the local offline store (SQLite/ObjectBox, per device) and the cloud store (PostgreSQL), with the same logical schema — this symmetry is what makes the Sync Engine's incremental push/pull straightforward.

Entity	Purpose
staff	Master record for every employee of a property (incl. the Owner).
roles	Defines a role (system-seeded or custom) scoped to a property.
role_permissions	Module-level CRUD/approve/export permission grants per role.
staff_attendance	One row per staff per calendar date.
leave_types	Lookup of leave categories (Casual, Sick, Paid, Unpaid, Emergency).
staff_leave	Leave applications and their approval lifecycle.
staff_salary	Monthly payroll record per staff.
staff_performance	Periodic performance review entries.
staff_tasks	Task assignment & tracking, optionally linked to another module's record.
staff_documents	ID proofs and HR documents with verification status.
staff_sessions	Login/logout session history, tied to a registered device.
staff_audit_log	Immutable log of every sensitive staff-related action.

Entity Relationship Summary (textual ER)

properties (external) 1—* staff | staff *—1 roles | roles 1—* role_permissions | staff 1—* staff_attendance | staff 1—* staff_leave *—1 leave_types | staff 1—* staff_salary | staff 1—* staff_performance | staff 1—* staff_tasks (as assignee) | staff 1—* staff_documents | staff 1—* staff_sessions *—1 devices (external) | staff 1—* staff_audit_log

7. Database Tables — Full Schema

Column key: **PK** = Primary Key, **FK** = Foreign Key, **UQ** = Unique, **NN** = Not Null. All primary keys are client-generatable UUIDs (v4) to remain conflict-safe under offline-first, multi-device writes.

7.1 staff

Master employee record.

Column	Type	Key	Notes
staff_id	UUID	PK	Client-generated identifier.
property_id	UUID	FK, NN	→ properties.property_id (external module).
employee_code	TEXT	UQ*	Unique per property; auto-generated or manual.
full_name	TEXT	NN	
mobile_number	TEXT	UQ*, NN	Unique per property; used for WhatsApp notifications.
email	TEXT		Optional.
gender	TEXT		
date_of_birth	DATE		
photo_url	TEXT		Object storage reference.
role_id	UUID	FK, NN	→ roles.role_id.
date_of_joining	DATE	NN	
date_of_leaving	DATE		Null while active.
employment_status	TEXT	NN	Active / Inactive / Suspended / Terminated.
address	TEXT		
emergency_contact_name	TEXT		
emergency_contact_number	TEXT		
biometric_id	TEXT		Local biometric template reference.
pin_hash	TEXT		Hashed PIN fallback login.
is_primary_owner	BOOLEAN	UQ*	Only one TRUE row per property_id.
created_by / updated_by	UUID	FK	Self-reference → staff.staff_id.
created_at / updated_at	DATETIME	NN	
version	INTEGER	NN	Optimistic concurrency counter, default 1.
sync_status	TEXT	NN	synced / pending / conflict.
last_synced_at	DATETIME		
is_deleted	BOOLEAN	NN	Soft delete flag, default false.

7.2 roles

Role catalogue, system-seeded or property-custom.

Column	Type	Key	Notes
role_id	UUID	PK	
property_id	UUID	FK	Null for global system roles.
role_name	TEXT	NN	e.g. Owner, Manager, Reception, Housekeeping.
is_system_role	BOOLEAN	NN	True for the 10 BRD-defined roles.
description	TEXT		
created_at / updated_at	DATETIME	NN	

7.3 role_permissions

Module-level permission grants per role.

Column	Type	Key	Notes
permission_id	UUID	PK	
role_id	UUID	FK, NN	→ roles.role_id.
module_name	TEXT	NN	Bookings / Check-in / Payments / Expenses / Reports / Room Pricing / Discount / Users / Inventory / Restaurant / Settings, etc.
can_view	BOOLEAN	NN	Default false.
can_create	BOOLEAN	NN	Default false.
can_edit	BOOLEAN	NN	Default false.
can_delete	BOOLEAN	NN	Default false.
can_export	BOOLEAN	NN	Default false.
can_approve	BOOLEAN	NN	Refunds, leave approvals, etc.
UQ (role_id, module_name)	—	UQ	One permission row per role per module.

7.4 staff_attendance

One attendance record per staff, per day.

Column	Type	Key	Notes
attendance_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
property_id	UUID	FK, NN	Denormalized for fast offline queries.
attendance_date	DATE	NN	
check_in_time	DATETIME		
check_out_time	DATETIME		
status	TEXT	NN	Present / Absent / Half-Day / On-Leave / Holiday.
check_in_method	TEXT		Biometric / PIN / Manual / Face.

Column	Type	Key	Notes
check_in_lat / lng	REAL		Optional geo-stamp.
marked_by	UUID	FK	→ staff.staff_id, for manual entries.
remarks	TEXT		
created_at / updated_at	DATETIME	NN	
sync_status	TEXT	NN	
UQ (staff_id, attendance_date)	—	UQ	Prevents duplicate daily entries.

7.5 leave_types

Lookup of leave categories.

Column	Type	Key	Notes
leave_type_id	UUID	PK	
property_id	UUID	FK	Null = global default type.
name	TEXT	NN	Casual / Sick / Paid / Unpaid / Emergency.
max_days_per_year	INTEGER		
is_paid	BOOLEAN	NN	

7.6 staff_leave

Leave applications and approvals.

Column	Type	Key	Notes
leave_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
leave_type_id	UUID	FK, NN	→ leave_types.leave_type_id.
from_date / to_date	DATE	NN	CHECK (from_date <= to_date).
total_days	INTEGER	NN	Derived from date range.
reason	TEXT		
status	TEXT	NN	Pending / Approved / Rejected / Cancelled.
applied_on	DATETIME	NN	
approved_by	UUID	FK	→ staff.staff_id.
approved_on	DATETIME		
rejection_reason	TEXT		
sync_status	TEXT	NN	

7.7 staff_salary

Monthly payroll record.

Column	Type	Key	Notes
salary_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
salary_month	INTEGER	NN	1–12.
salary_year	INTEGER	NN	
basic_salary	REAL	NN	
allowances	REAL	NN	Default 0.
overtime_amount	REAL	NN	Default 0.
deductions	REAL	NN	Default 0.
advance_deducted	REAL	NN	Default 0.
net_salary	REAL	NN	Computed field.
payment_status	TEXT	NN	Pending / Partially Paid / Paid.
payment_date	DATETIME		
payment_mode	TEXT		Cash / Bank Transfer / UPI / Cheque.
generated_by	UUID	FK	→ staff.staff_id.
remarks	TEXT		
UQ (staff_id, salary_month, salary_year)	—	UQ	One payroll row per staff per month.

7.8 staff_performance

Periodic performance reviews.

Column	Type	Key	Notes
performance_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
review_period_start / end	DATE	NN	
rating	REAL	NN	CHECK (rating BETWEEN 0 AND 5).
attendance_score	REAL		Derived, cached at review time.
task_completion_score	REAL		Derived, cached at review time.
remarks	TEXT		
reviewed_by	UUID	FK, NN	→ staff.staff_id.
review_date	DATETIME	NN	

7.9 staff_tasks

Task assignment & tracking.

Column	Type	Key	Notes
task_id	UUID	PK	
staff_id	UUID	FK, NN	Assignee → staff.staff_id.
assigned_by	UUID	FK, NN	→ staff.staff_id.
property_id	UUID	FK, NN	
task_title	TEXT	NN	
task_description	TEXT		
related_module	TEXT		Housekeeping / Maintenance / Restaurant / Front-Desk / General.
related_record_id	UUID		e.g. room_id, complaint_id — polymorphic link.
priority	TEXT	NN	Low / Medium / High / Urgent.
status	TEXT	NN	Pending / In-Progress / Completed / Cancelled.
due_date	DATETIME		
completed_at	DATETIME		
completion_notes	TEXT		
completion_photo_url	TEXT		

7.10 staff_documents

HR documents & ID proofs.

Column	Type	Key	Notes
document_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
document_type	TEXT	NN	Aadhaar / PAN / Passport / Driving License / Police Verification / Offer Letter / Other.
document_number	TEXT		
file_url	TEXT		Object storage reference.
is_verified	BOOLEAN	NN	Default false.
verified_by	UUID	FK	→ staff.staff_id.
verified_on	DATETIME		

7.11 staff_sessions

Login/logout session & device binding.

Column	Type	Key	Notes
session_id	UUID	PK	
staff_id	UUID	FK, NN	→ staff.staff_id.
device_id	UUID	FK, NN	→ devices.device_id (external Device Management module).
login_time	DATETIME	NN	

Column	Type	Key	Notes
logout_time	DATETIME		
login_method	TEXT		Biometric / PIN / Password.
is_active	BOOLEAN	NN	Default true; forced false on Remote Device Control revoke.
ip_address	TEXT		Captured when online.

7.12 staff_audit_log

Immutable action log (no application-level delete).

Column	Type	Key	Notes
log_id	UUID	PK	
staff_id	UUID	FK	Actor performing the action; null for system jobs.
property_id	UUID	FK, NN	
action	TEXT	NN	Created / Updated / Deleted / Login / Logout / Approved / Rejected.
module_name	TEXT	NN	e.g. Staff, Attendance, Leave, Salary.
target_entity	TEXT	NN	Affected table name.
target_record_id	UUID	NN	
old_value / new_value	TEXT		JSON snapshots.
performed_at	DATETIME	NN	
is_immutable	BOOLEAN	NN	Always true; enforced by DB-role revoke on DELETE.

8. Database Relationships & Constraints Summary

From	References	Cardinality	Rule
staff.property_id	properties.property_id	Many-to-One	ON DELETE RESTRICT (external module).
staff.role_id	roles.role_id	Many-to-One	ON DELETE RESTRICT — a role in use cannot be deleted.
staff.created_by / updated_by	staff.staff_id	Self-referencing	Nullable, ON DELETE SET NULL.
role_permissions.role_id	roles.role_id	Many-to-One	ON DELETE CASCADE.
staff_attendance.staff_id	staff.staff_id	Many-to-One	ON DELETE CASCADE (soft-delete preferred).
staff_attendance.marked_by	staff.staff_id	Many-to-One	ON DELETE SET NULL.
staff_leave.staff_id	staff.staff_id	Many-to-One	ON DELETE CASCADE.
staff_leave.leave_type_id	leave_types.leave_type_id	Many-to-One	ON DELETE RESTRICT.
staff_leave.approved_by	staff.staff_id	Many-to-One	ON DELETE SET NULL.
staff_salary.staff_id	staff.staff_id	Many-to-One	ON DELETE RESTRICT (financial record retained).
staff_salary.generated_by	staff.staff_id	Many-to-One	ON DELETE SET NULL.
staff_performance.staff_id	staff.staff_id	Many-to-One	ON DELETE CASCADE.
staff_performance.reviewed_by	staff.staff_id	Many-to-One	ON DELETE SET NULL.
staff_tasks.staff_id	staff.staff_id	Many-to-One	ON DELETE SET NULL (task remains, unassigned).
staff_tasks.assigned_by	staff.staff_id	Many-to-One	ON DELETE SET NULL.
staff_documents.staff_id	staff.staff_id	Many-to-One	ON DELETE CASCADE.
staff_sessions.staff_id	staff.staff_id	Many-to-One	ON DELETE CASCADE.
staff_sessions.device_id	devices.device_id	Many-to-One	ON DELETE RESTRICT (external module).
staff_audit_log.staff_id	staff.staff_id	Many-to-One	ON DELETE SET NULL — log row is never removed.

Key Constraints

- Exactly one staff row per property may have `is_primary_owner = true` (partial unique index).
- `employee_code` and `mobile_number` are unique per `property_id`, not globally — the same phone number can belong to staff at two different properties in a multi-tenant SaaS.
- `staff_attendance` enforces one row per staff per date.
- `staff_salary` enforces one row per staff per (month, year).
- `staff_leave` enforces `from_date ≤ to_date` via a CHECK constraint.
- `staff_audit_log` has no DELETE grant at the database-role level — matching the BRD's 'Nothing can be deleted from the audit log' rule.
- All tables carry `sync_status`, `version` and timestamps to support the Incremental Sync + Conflict Resolution features from the Synchronization Engine (BRD Page 4).

9. Dashboards — Count, Use Cases & Fields

The Staff Management module requires **5 distinct dashboards**, mapped to the different actors in the BRD's User Hierarchy (Page 18) and Role Matrix (Page 32–33): Owner/Manager, Reception-level staff directory users, day-to-day attendance operators, staff themselves (self-service), and Pinesphere's Super Admin for cross-property oversight.

9.1 Owner / Manager — Staff Overview Dashboard

Primary Users: Owner, Manager

Use Case: Give the property owner or manager a single-glance operational view of the entire workforce, matching the style of the Owner Dashboard already defined in the BRD (Page 7).

Fields / Widgets:

Field / Widget	Description
Total Staff Count	Active headcount for the property.
Active / Inactive Staff	Split count by employment_status.
Present Today / Absent Today	Derived from staff_attendance for current date.
On Leave Today	Count of staff with status On-Leave today.
Pending Leave Requests	Count of staff_leave rows with status Pending, tap-through to approve/reject.
Pending / Overdue Tasks	Count of staff_tasks not completed; overdue highlighted in red.
Today's Attendance %	Present ÷ Total Active Staff.
Upcoming Salary Due	Count of staff with no Paid salary row for the current month.
Performance Alerts	Staff with latest rating below a configurable threshold.
Role-wise Staff Distribution	Chart — staff count grouped by role.
Recently Added Staff	Last 5 onboarded staff with joining date.
Quick Actions	Add Staff, Approve Leave, Assign Task, Generate Salary.

9.2 Staff Directory Dashboard

Primary Users: Owner, Manager, Accountant (view-only where applicable)

Use Case: Browse, search and manage the full staff list — the operational hub for HR actions, feeding the Owner's 'Create Staff' / 'Manage Users' capability from Page 27.

Fields / Widgets:

Field / Widget	Description
Photo, Name, Employee Code	Identity summary per row.
Role Badge	Color-coded role tag.
Mobile Number	Tap to call / WhatsApp.
Employment Status	Active / Inactive / Suspended chip.
Date of Joining	

Field / Widget	Description
Last Attendance	Most recent check-in date/time.
Filters	By Role, Status, Date of Joining range.
Search Bar	By name, employee code, mobile number.
Row Actions	View Profile, Edit, Deactivate, Reset PIN, View Documents.

9.3 Attendance Dashboard

Primary Users: Owner, Manager, Reception (for manual marking)

Use Case: Track daily and monthly attendance across all staff, and let an authorized user record a manual punch for staff who could not use biometric/PIN.

Fields / Widgets:

Field / Widget	Description
Date Selector	Single date or month-range view.
Attendance List	Staff name, check-in time, check-out time, status, method icon.
Daily Counters	Present / Absent / Half-Day / On-Leave totals.
Monthly Calendar View	Per-staff calendar heat-map of attendance status.
Late Arrivals Count	Staff checked in after configured shift start time.
Manual Mark Action	Mark Present/Absent/Half-Day for a staff member with remarks.
Export	Export attendance sheet for a date range (Owner/Accountant only).

9.4 Staff Self-Service Dashboard

Primary Users: Any logged-in staff member (mobile)

Use Case: Each staff member's personal home screen after login — works fully offline, syncs later.

Fields / Widgets:

Field / Widget	Description
Punch In / Punch Out Button	Primary action, large tap target.
This Month's Attendance	Present / Absent / Leave day counts.
Leave Balance	Remaining days per leave type; Apply for Leave action.
My Tasks	Pending, In-Progress, Completed task list with due dates.
My Salary History	Past salary slips with payment status.
My Performance Rating	Latest rating + trend, if visible to the role.
My Profile & Documents	View/update personal details and uploaded documents.

9.5 Super Admin — Staff Analytics Dashboard (Cross-Property)

Primary Users: Pinesphere Super Admin, Support Team

Use Case: Gives Pinesphere operational visibility into staff usage across every customer property, extending the Super Admin Dashboard already defined in the BRD (Page 18–19) with staff-specific analytics.

Fields / Widgets:

Field / Widget	Description
Total Staff Across All Properties	Global headcount.
Staff Growth Trend	New staff onboarded over time, chart.
Role Distribution (Global)	Aggregate staff count by role across all properties.
Average Staff per Property	By subscription plan tier (Starter / Basic / Standard / Professional / Enterprise).
Properties with Sync Issues	Properties whose staff data has pending/conflict sync_status.
Device–Staff Mapping Anomalies	Staff sessions active on unregistered/unapproved devices.
Top Properties by Staff Count	Ranked list, links to Property Management screen.

10. Security & Compliance Considerations

- **Local database encryption** for the SQLite/ObjectBox store, per BRD Page 15.
- **Biometric login with PIN fallback** for staff punch-in and app access.
- **Role-based permissions** enforced both client-side (UX gating) and server-side (API authorization) — never trust the client alone.
- **Device registration** — a staff member can only log in from a device approved under that property (Device Management module).
- **Remote deactivation** — Super Admin or Owner can revoke a staff member's session; local data is erased from that device on next reconnect if the staff is terminated.
- **Audit trail immutability** — enforced at the database role level (REVOKE DELETE on staff_audit_log).
- **Encrypted sync** for all staff payloads travelling between device and cloud.
- **PII minimization** — Aadhaar/Passport numbers stored in staff_documents only, not duplicated into the staff master row.

11. Scalability & Flexibility Notes

- **Data-driven roles** — roles and permissions live in tables, not code, so a new role (e.g. 'Night Auditor') can be added per property without an app release.
- **Multi-tenant safe uniqueness** — all uniqueness constraints (mobile number, employee code) are scoped by property_id, so the same schema serves Starter homestays and Enterprise multi-property chains alike.
- **Polymorphic task linkage** — staff_tasks.related_module / related_record_id lets the task engine attach to any future module (e.g. Inventory restock task) without schema changes.
- **Plan-aware limits** — Maximum Staff Users per subscription plan (BRD Page 23) can be enforced by a simple count check against the staff table at creation time, without new tables.
- **Derived, not duplicated, metrics** — performance scores are computed from attendance/task tables at read time, keeping the schema lean and avoiding stale cached numbers.
- **Sync-ready by design** — every table already carries the version/sync_status/timestamps needed for incremental sync, so this module plugs into the existing Synchronization Engine without special-casing.
- **Backend framework** — FastAPI's Pydantic models map 1:1 to these tables, keeping request/response validation and the offline payload contract in sync as the schema evolves.

12. Appendix — Role Permission Matrix (BRD Reference)

Reproduced and extended from the BRD's Recommended Role Matrix (Pages 32–33), focused on modules relevant to Staff Management. 'View' = read-only, 'Full' = all actions, '—' = no access.

Module	Super Admin	Owner	Manager	Reception	Accountant	Staff (Self)
Staff / User Management	Full	Full	Limited	—	—	—
Attendance	Full	Full	Own	Own	Own	Own
Leave Approval	Full	Full	—	Apply	Apply	Apply
Task Assignment	Full	Full	Assign/Own	Own	Own	Own
Salary / Payroll	Full	Full	—	—	Full (Accountant)	Own
Performance Review	Full	Full	Limited	—	—	Own
Audit Logs	Full	Property Only	Limited	Own Actions	Own Actions	—

This blueprint is derived entirely from PineStay — BRD.pdf and is scoped to the Staff Management module for V1. All entity, workflow and dashboard decisions map back to explicit BRD sections cited throughout this document (Pages 7, 10, 15, 18, 26–29, 31–33).